



EXAM PAPERS PRACTICE

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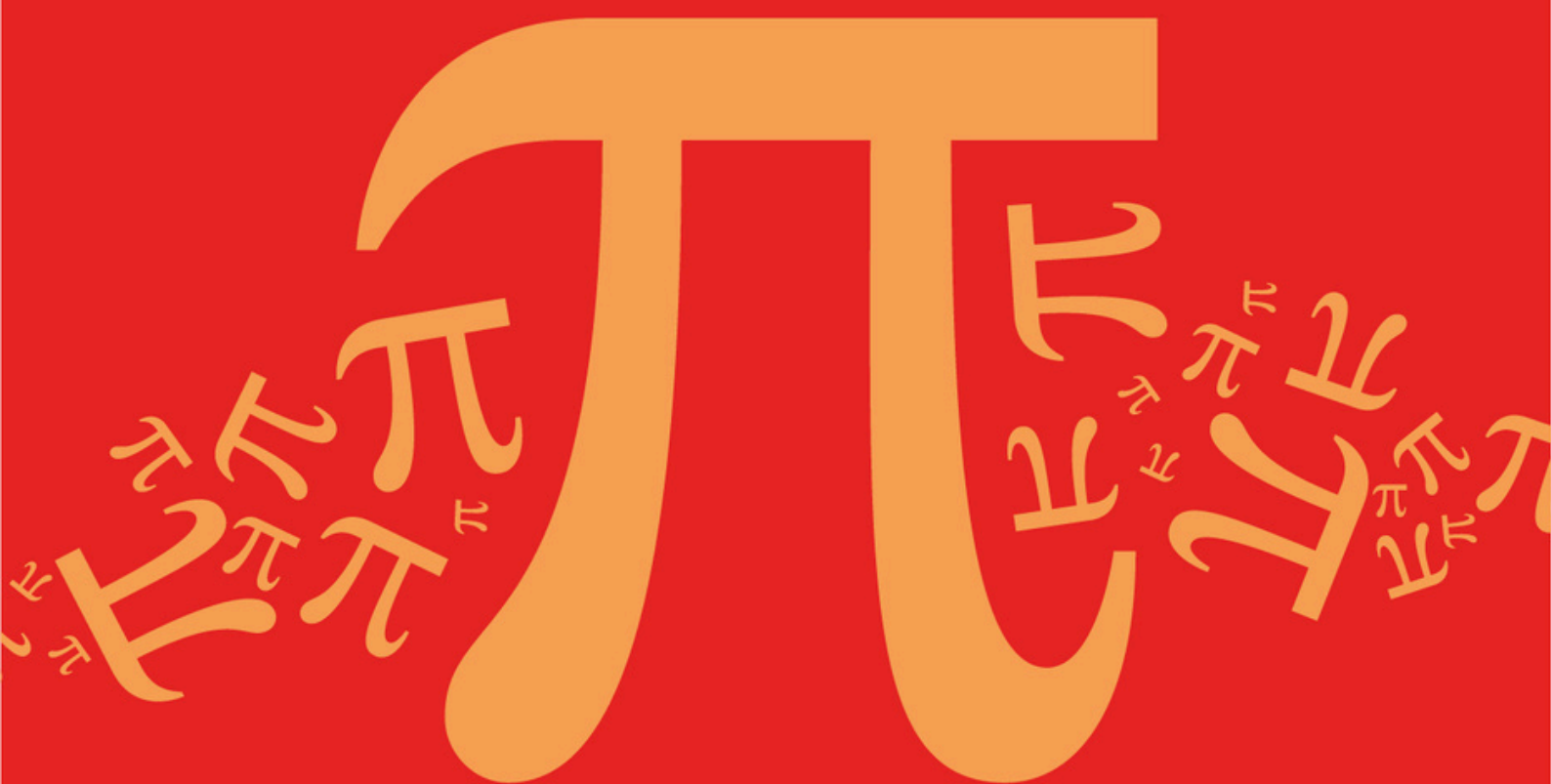
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Detailed mark scheme

Suitable for all boards

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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D6: Use Standard Maclaurin Series

Topic Questions

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Question Paper

Topic

D: Further Algebra and Functions

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D6: Use Standard Maclaurin Series

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Which of the following functions has the fourth term $-\frac{1}{720}x^6$ in its Maclaurin series expansion?

Circle your answer.

$\sin x$

$\cos x$

e^x

$\ln(1+x)$

(Total 1 mark)

Q2.

- (a) Write down the expansion of e^{3x} in ascending powers of x up to and including the term in x^2 .

(1)

- (b) Hence, or otherwise, find the term in x^2 in the expansion, in ascending powers of

x , of $e^{3x}(1+2x)^{-\frac{3}{2}}$.

(4)

(Total 5 marks)

Q3.

- (a) Write down the expansions in ascending powers of x up to and including the term in x^3 of:

(i) $\cos x + \sin x$,

(1)

(ii) $\ln(1+3x)$.

(1)

- (b) It is given that $y = e^{\tan x}$.

(i) Find $\frac{dy}{dx}$ and show that $\frac{d^2y}{dx^2} = (1+\tan x)^2 \frac{dy}{dx}$

(5)

(ii) Find the value of $\frac{d^3y}{dx^3}$ when $x = 0$.

(2)

- (iii) Hence, by using Maclaurin's theorem, show that the first four terms in the expansion, in ascending powers of x , of $e^{\tan x}$ are

$$1 + x + \frac{1}{2}x^2 + \frac{1}{2}x^3$$

(2)

(c) Find

$$\lim_{x \rightarrow 0} \left[\frac{e^{\tan x} - (\cos x + \sin x)}{x \ln(1 + 3x)} \right]$$

(3)

(Total 14 marks)

Q4.

The function f is defined by $f(x) = e^{2x}(1 + 3x)^{-\frac{2}{3}}$.

- (a) (i) Use the series expansion for e^x to write down the first four terms in the series expansion of e^{2x} .

(2)

- (ii) Use the binomial series expansion of $(1 + 3x)^{-\frac{2}{3}}$ and your answer to part (a)(i) to show that the first three non-zero terms in the series expansion of $f(x)$ are $1 + 3x^2 - 6x^3$.

(5)

- (b) (i) Given that $y = \ln(1 + 2 \sin x)$, find $\frac{d^2 y}{dx^2}$.

(4)

- (ii) By using Maclaurin's theorem, show that, for small values of x ,

$$\ln(1 + 2 \sin x) \approx 2x - 2x^2$$

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(2)

(c) Find

$$\lim_{x \rightarrow 0} \frac{1 - f(x)}{x \ln(1 + 2 \sin x)}$$

(3)

(Total 16 marks)

Q5.

- (a) The function f is defined by

$$f(x) = \ln(1 + e^x)$$

Use Maclaurin's theorem to show that when $f(x)$ is expanded in ascending powers of x :

- (i) the first three terms are

$$\ln 2 + \frac{1}{2}x + \frac{1}{8}x^2$$

(6)

(ii) the coefficient of x^3 is zero.

(3)

(b) Hence write down the first two non-zero terms in the expansion, in ascending powers of x , of $\ln\left(\frac{1+e^x}{2}\right)$.

(1)

(c) Use the series expansion

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \dots$$

to write down the first three terms in the expansion, in ascending powers of x , of $\ln\left(1 - \frac{x}{2}\right)$.

(1)

(d) Use your answers to parts (b) and (c) to find

$$\lim_{x \rightarrow 0} \left[\frac{\ln\left(\frac{1+e^x}{2}\right) + \ln\left(1 - \frac{x}{2}\right)}{x - \sin x} \right]$$

(4)

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